Consider the following two reactions.

Reaction 1 : $4\ Fe + 3O_2 \rightarrow 2\ Fe_2O_3$

Reaction 2 : $H_2S + Cl_2 \rightarrow 2\ HCl + S$

Both these reactions are oxidation reactions as per the classical concept.

In the first reaction which is responsible for the rusting of iron, the oxygen adds on to the metal, iron. In the second reaction, hydrogen is removed from Hydrogen sulphide (H_2S). Identity which species gets reduced.

Consider the following two reactions in which the removal of oxygen and addition of hydrogen take place respectively. These reactions are called reduction reactions as per the classical concept.

$CuO + C \rightarrow Cu + CO$ (Removal of oxygen from cupric oxide)

$S + H_2 \rightarrow H_2S$ (Addition of hydrogen to sulphur).

Oxidation-reduction reactions i.e. redox reactions are not always associated with oxygen or hydrogen. In such cases, the process can be explained on the basis of electrons. The reaction involving loss of electron is termed oxidation and gain of electron is termed reduction.

For example,

$Fe^{2+} \rightarrow Fe^{3+} + e^-$ (loss of electron-oxidation).

$Cu^{2+} + 2e^- \rightarrow Cu$ (gain of electron-reduction)

Redox reactions can be better explained using oxidation numbers.

1.8.1 Oxidation Number:

It is defined as the imaginary charge left on the atom when all other atoms of the compound have been removed in their usual oxidation states that are assigned according to set of rules. A term that is often used interchangeably with oxidation number is oxidation state

1) The oxidation state of a free element (i.e. in its uncombined state) is zero.

Example : each atom in H_2, Cl_2, Na, S_8 have the oxidation number of zero.

2) For a monatomic ion, the oxidation state is equal to the net charge on the ion.

Example : The oxidation number of sodium in Na^+ is +1.

The oxidation number of chlorine in Cl^- is –1.

3) The algebric sum of oxidation states of all atoms in a molecule is equal to zero, while in ions, it is equal to the net charge on the ion.

Example:

In H_2SO_4; 2 × (oxidation number of hydrogen) + (oxidation number of S) + 4 (oxidation number of oxygen) = 0.

In SO_4^{2-}; (oxidation number of S) + 4 (oxidation number of oxygen) = –2.

4) Hydrogen has an oxidation number of +1 in all its compounds except in metal hydrides where it has –1 value.

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